Chuan Qin

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Google Scholar: https://scholar.google.com/citations?user=17NMHvUAAAAJ

Metrics: 86 Citations, h-index=2, i10-index=2

EDUCATION

Lanzhou University Sept. 2021 - Present (Expected 2027)
Computer Application Technology, PhD Candidate (Master-PhD Continuation)

Harbin Institute of Technology Sept. 2017 - Jun. 2021
Naval Architecture and Ocean Engineering, Bachelor

RESEARCH INTERESTS

Neural Network Optimization; Generalization; Robot Control

PUBLICATIONS

[1] C. Qin and L. Jin, "A survey of large-model-based AI agents," ***Tsinghua Science and Technology***, In Press.

[2] C. Qin et al., "Long short-term memory with activation on gradient," ***Neural Networks***, vol. 164, pp. 135-145, 2023. (SCI Q1, IF=6.3, Cited 76)

[3] C. Qin et al., "Adaptive gradient activation for federated learning on consumer electronic devices," ***IEEE Transactions on Consumer Electronics***, 2025. (IF=10.9, Cited 10)

[4] C. Qin et al., "SAM-LA: Efficient Sharpness-Aware Minimization Training via Loss Acceleration," ***IEEE/CAA Journal of Automatica Sinica***. (SCI Q1, IF=18.3, Accept w/ Minor Revision)

[5] C. Qin et al., "Adaptive gradient reshaping boosting generalization," ***IEEE Transactions on Neural Networks and Learning Systems***. (SCI Q1, Minor Revision)

[6] C. Qin and L. Jin, "Collaborative neurodynamic optimization with gradient norm penalty for generalization improvement," ***IEEE Transactions on Cybernetics***. (SCI Q1, IF=10.5, Reject & Resubmit)

[7] C. Qin and L. Jin, "Anti-Perturbation Solution of Dynamic Multilinear Quaternion Tensor Equations With Application to Manipulators," ***CAAI Transactions on Intelligence Technology***. (SCI Q1, Major Revision)

[8] C. Qin and L. Jin, "Fuzzy Sign-Based Collaborative Neurodynamic Optimization With Application to Robots," ***IEEE Transactions on Fuzzy Systems***. (SCI Q1, Under Review)

[9] C. Qin et al., "SAM-L: Switching SAM with loss gap for efficient training," ***NeurIPS 2026***. (Under Review)

[10] C. Qin et al., "On Ternary Duality of Dynamical Systems, Information Theory, and Deep Learning," ***PANS***. (Submitted)

[11] C. Qin et al., "Consistency is key of fitting and degradation alleviating for deep neural networks," ***IEEE Transactions on Neural Networks and Learning Systems***. (Under Review)

[12] C. Qin et al., "Quantum-Inspired Collaborative Neurodynamic Optimization for Traffic Flow Prediction," ***IEEE Transactions on Consumer Electronics***. (Under Review)

[13] C. Qin et al., "Understanding Residual Neural Networks from Sampling Theorem Perspective," ***IEEE Transactions on Emerging Topics in Computational Intelligence***. (Under Review)

RESEARCH PROJECTS (6)

[1]Research on Differential Equation Interpretation of Generalization and Robustness of Large Models, CCF-Huawei Populus Fund, CNY 110K, 2024.10-2025.10.
Contributions: Led proposal writing, mid-term review, and final project closure; designed research plans and coordinated team collaboration.

[2]Stability, Convergence and Optimization Strategies of Large Models via Neural Differential Equations, CCF-Baidu Pinecone Fund, CNY 150K, 2023.11-2024.10.

Contributions: Full-cycle project management; independently conducted core algorithm R&D and coordinated team progress.

[3]Deep Neural Network Structure Design and Optimizer Construction via Differential Equations Based on MindSpore, CAAI-Huawei MindSpore Fund, CNY 90K, 2022.11-2023.11.

Contributions: Led proposal and closure; participated in core algorithm design.

[4]Neural ODE Basic Theory, Learning Algorithms and Applications, Central University Basic Research Fund, CNY 400K, 2023.06-2026.06.

Contributions: Participated in algorithm design and experiments.

[5]Stability and Optimization of Large Language Models for Tibetan Information Processing, State Key Laboratory Open Project, CNY 40K, 2024.1-2025.12.

Contributions: Participated in algorithm design and experiments.

[6]Gansu Provincial Graduate Student "Innovation Star" Project, Gansu Provincial Department of Education, CNY 10K, 2026.1-2026.12.

Contributions: Principal Investigator (PI), led project application, execution, and closure.

HONORS & AWARDS

- First Prize, Lanzhou University Academic Annual Conference Paper
- First Prize, Academic Scholarship
- Outstanding Graduate Student, Lanzhou University

OTHERS

- Half Marathon Finisher: 2024 Lanzhou Marathon (Net: 01:42:38), 2024 Yingshan Marathon (Net: 01:53:39)